Mahmoud Alyosify

MLOps & Cloud AI Engineer | Model Deployment, Pipelines & Cost-Efficient Inference
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PROFESSIONAL SUMMARY

MLOps & Cloud AI Engineer focused on the gap between a trained model and a system that runs reliably and affordably. Currently an **AI Engineer Intern at Injaz Tech**, taking an agentic RAG system to a **FastAPI** service with **Docker**, **GitHub Actions** CI, an automated evaluation gate, Prometheus metrics and per-node cost reporting. Deployed **fine-tuned LLMs and vision encoders** to **AWS** via **ONNX, Docker and FastAPI**; ran master's-thesis experiments as **deterministic, reproducible benchmarks on a pinned NVIDIA A40** (pinned model revisions, every number recomputable from stored answers); and trained **deep-RL agents that provision cloud resources against an explicit cost–latency budget**. Holds **three AWS Academy credentials** (ML Foundations, ML for NLP, Data Engineering). Completing an **MSc in Artificial Intelligence at Queen's University, Canada** (expected **Nov 2026**) on a **Digilians Presidential Scholarship**, with prior production-systems experience on **national-scale toll infrastructure**.

EDUCATION

MSc in Artificial Intelligence — *Sep 2025 – Nov 2026*
Queen's University, Kingston, Ontario, Canada — Expected Nov 2026 · GPA: 4.30 / 4.30

- **Digilians Presidential Scholarship** — competitive national AI graduate fellowship.
- Coursework: **Cloud Computing, End-to-End ML Pipelines, Deep Learning, Large Language Models, Reinforcement Learning**.

B.Sc. Computer & Information Science (Bioinformatics) — *Sep 2019 – Jul 2023*
Assiut University, Egypt — GPA: 3.53 / 4.00

- Graduation Project: **Vitalism Solution** — **Grade A+**; **Semi-Finalist, Microsoft Imagine Cup 2023**.

TECHNICAL SKILLS

Cloud & Infrastructure	AWS (GPU Instances, ML & Data Services), Docker, Docker Compose, CI/CD (GitHub Actions), Linux, Bash, Infrastructure Cost Optimisation, Autoscaling Strategies, Self-Hosted Model Serving
Model Serving & Deployment	ONNX, FastAPI, Streamlit, FAISS, ChromaDB, Containerised Inference, Batch & Interactive Serving, SSE Streaming, Health & Readiness Probes, Response Caching, API Design
Reliability & Measurement	Deterministic & Reproducible Pipelines, Pinned Environments and Seeds, Latency Profiling (Per-Token, Per-Answer), Throughput Benchmarking, GPU-Time Accounting, Baseline Comparison, Retries & Circuit Breakers, Structured Logging, Prometheus Metrics, CI Evaluation Gates, pytest
Data Engineering	ETL Pipelines, PySpark, Pandas, NumPy, SQL, Data Validation, MySQL, MongoDB, SQL Server
ML & AI	PyTorch, TensorFlow, Scikit-learn, Hugging Face, LangChain, LangGraph, RAG, Fine-Tuning (LoRA, QLoRA, PEFT), Reinforcement Learning (PPO, DQN), Optuna
Programming	Python, SQL, C#, C++, C, Go, JavaScript, Bash

PROFESSIONAL EXPERIENCE

AI Engineer Intern — *Oct 2026 – Present*
Injaz Tech — Injaz Group (formerly Fingerprint Consultancy) — Maadi, Cairo, Egypt

- Took an agentic RAG system to a **FastAPI** service: **Docker** / docker-compose, typed configuration, health and readiness probes, SSE streaming.
- Built **GitHub Actions** CI (lint + 136 tests, nightly evaluation gate), **Prometheus** metrics, per-node cost reporting, retries and a circuit breaker.

Part-Time Instructor — Machine Learning & AI for Business — *Jul 2025 – Present*
National Telecommunication Institute (NTI) — Egypt

- Teach **Machine Learning** (90-hour bootcamps) and **AI for Business** modules in NTI's Summer Internships Program, on separate engagements.

AI in Cybersecurity Intern — *Jun 2025 – Jul 2025*
e& Egypt — via NTI (On-the-Job Training) — Cairo, Egypt

- Applied AI-driven security tooling across threat intelligence, digital forensics and enterprise security analysis.

Systems Engineer (Military Service) — *Feb 2024 – Mar 2025*
National Company for Roads Building & Development — Cairo, Egypt

- Supported production backend systems for national toll infrastructure serving **50%+ of Egyptian vehicles** across **70%+ of national roads**, where downtime had immediate operational cost.
- Debugged and optimised real-time transportation software and supported legacy applications to maintain continuity; co-developed an internal data-management and reporting application in **C#, .NET, Entity Framework, SQL Server** and **Crystal Reports**.

Online Instructor — AI, Algorithms & Data Science *2020 – Present*
Udemy & YouTube Remote

- Built Arabic-language technical curricula reaching **10,000+ learners** and **700K+ YouTube views**.

SELECTED DEPLOYMENT & INFRASTRUCTURE PROJECTS

RL-Cloud-Autoscaler — Learned Cloud Resource Provisioning *2026*
Python · PPO · DQN · Gymnasium · Cloud Cost Optimisation

- Built a **custom Gymnasium environment** simulating cloud workload dynamics, then trained **PPO** and **DQN** agents to make provisioning decisions in real time.
- Optimised an explicit **cost–latency reward** and benchmarked against **static allocation** and **reactive threshold-based autoscaling** — the policies teams actually run — to quantify real savings rather than simulated ones.

MinimalLM: Verbosity-Aware Decoding for Token-Efficient Language Model Inference *2026*
MSc Thesis (CISC 898), Queen's University · NVIDIA A40 GPU · Reproducible Evaluation

- Fewer generated tokens are a real saving only when the answer is still complete. **MinimalLM gives both, without training. 47.6%** fewer generated tokens and **19–68%** faster answers on three open LLMs, with **no answer cut off**; time per token within 0.8% of the baseline, and in fixed batches GPU time falls **5–55%**.
- Reproducible by design: pinned model revisions, greedy decoding, a pinned **NVIDIA A40**, **29** GPU invariant tests, 27 unit tests, and a notebook that recomputes every table and number from the stored answers and re-runs all six systems.

Horus-OSINT — Fine-Tuned LLM Deployed on AWS *2026*
Meta-Llama-3-8B · Fine-Tuning · AWS · Self-Hosted Serving

- Owned the full path from a **159,826-record** dataset through fine-tuning to a running **AWS-hosted** interactive service — data preparation, training, packaging and deployment.

SimCLR-Vision-SSL — ONNX Export & Vector-Search Serving *2026*
PyTorch · ONNX · FAISS · Retrieval

- Exported a trained encoder to **ONNX** and served embeddings through a **FAISS** index — converting a research model into a framework-independent, deployable retrieval component.

Additional Projects

- **HORUS Sentinel** — multi-agent platform with a self-hosted fine-tuned LLM and knowledge-graph state layer running continuously.
- **Automated Multimodal Agent** — multi-stage automation pipeline chaining extraction, generation and synthesis services.
- **Also:** SCRAP · AudioShield (Streamlit-served CNN) · Human vs. Agentic Pull Requests · Vitalism Solution.

CERTIFICATIONS & AWARDS

AWS Academy — Data Engineering
Machine Learning Specialization — DeepLearning.AI
Full Stack Diploma — .NET & Angular (Route Academy)
1st Place — Smart Cities Hackathon 2022
AWS Academy — ML Foundations, ML for NLP
HCIP – AI (Huawei)
Microsoft Certified: Data Analyst Associate
Semi-Finalist — Microsoft Imagine Cup 2023

LANGUAGES

- **Arabic:** Native · **English:** Professional Working Proficiency.